LETLEARNTOGETHER · FRONTEND DESIGN SPECIFICATION

One platform.
One consistent interface.

Implemented routes, visual language, and role-based screens — from the home page to the last administrative interface.



Design reference · Based on the current frontend implementation · 2026

LetLearnTogether frontend design specification

This document records the interfaces that are implemented in the frontend. It is an implementation reference, not a proposal for a new visual design. When this document and the running interface differ, update the document from the code (or change the code and this document together).

1. Design source of truth

- Application entry and route definitions: `frontend/src/App.jsx`
- Public navigation: `frontend/src/components/layout/Navbar.jsx`
- Signed-in navigation and responsive workspace shell:

`frontend/src/components/layout/AuthenticatedLayout.jsx`

- Shared colors and base typography: `frontend/src/index.css`
- Main public-page and library styles: `frontend/src/App.css`
- Interface-specific styles: the page/component CSS files imported by each page.

The application uses React Router. The same route can render inside the public shell or the signed-in workspace shell depending on authentication. The route tables below distinguish the underlying interface from the shell around it.

2. Shared visual language

Element	Implemented design
Typeface	Inter, Segoe UI, Roboto, Helvetica, Arial, sans-serif
Primary blue	`#2459a4`
Dark brand blue	`#173b6c`
Body text	`#17243a`
Muted text	`#748197`
Page background	`#f7f9fc`; signed-in workspace background `#f4f7fb`
Surfaces and borders	White surfaces; pale blue/gray borders such as `#e7ecf3`
Controls	Rounded buttons and fields, visible keyboard-focus outlines, blue primary actions
Motion	Short page, section, and card entrance/hover transitions; animations are disabled or reduced when reduced motion is requested

Public shell

Unauthenticated pages use a white navigation bar with the LetLearnTogether mark/name, Home, Document Library, About, install-app control, Login, and Register. At widths of 700px and below, the navigation collapses behind a menu button. Public pages also show the platform notice when one is configured. The home and About pages finish with the site footer.

Signed-in shell

Authenticated routes render inside a workspace layout with a sticky header, brand, install-app control, user name/role, sign-out action, and a left-side workspace navigation. Navigation links depend on the signed-in role. Students also have a bottom mobile navigation for Library, Quizzes, Upload, and Me. The workspace provides a light/dark theme toggle; its choice is stored in `localStorage`. The sidebar becomes a responsive, dismissible menu on smaller screens.

The platform notice remains above either shell when enabled. Loading the session/settings and settings-load errors are application states, not separate navigation destinations.

3. Home and public interfaces

Order	Route	Interface and exact contents
1	`/` (signed out)	**Home.** Public navigation and optional platform notice; hero with "YOUR LEARNING LIBRARY," "Learn more. Achieve more.", a short resource description, and an "Explore documents" link to `/library`; the supplied study-group photograph with a "Verified resources" badge; public footer. The Document Library interface is not displayed on the home page.
2	`/library` (signed out or signed in)	**Document Library.** Dedicated searchable/filterable library page with course folders, announcements when a course is selected, reset filters, verified-resource notice, document cards, and pagination. Signed-in visitors see the authenticated shell.
3	`/courses/:courseCode` and deeper `/courses/:courseCode/:year`, `/courses/:courseCode/:year/:semester`, `/courses/:courseCode/:year/:semester/:documentType`, and `/courses/:courseCode/:year/:semester/:documentType/:lecturer`	**Course-folder library.** The same library interface, initialized from the route's course/year/semester/type/lecturer folders. A selected course can show its public announcements. A document-type folder without a lecturer prompts the visitor to select the lecturer folder.
4	`/about`	**About LetLearnTogether.** About hero and calls to explore the library or join; who-we-are section; mission and vision cards; six offering cards; values and audience panels; why section and closing quote; public footer. Signed-in visitors receive this content inside their workspace shell.
5	`/verify`	**Verify a document.** Verification ID entry form and submit action; result area is initially empty.
6	`/verify/:token`	**Document verification result.** The same verification form, prefilled from the token route, plus the lookup status and either verification error or result with status, document/course/type/approval details and QR code; link back to the library.
7	`/quizzes`	**Published quiz catalogue.** Search, course/year/semester filters, clear filters, result count, quiz cards and pagination; empty and load-error states. This route is accessible without signing in.
8	`/login`	**Sign in.** Logo, "Welcome back," email and password fields, show/hide password, sign-in action, inline validation/API feedback, and create-account link. After successful sign-in, the user is taken to their role dashboard or allowed requested destination.
9	`/register`	**Create account.** Full name, email, password, confirm password, show-password checkbox, inline validation, create-account action, and sign-in link. If registration is disabled in platform settings, the form is replaced by "Registration is paused" and a sign-in link.

Home hero image

- Asset: `frontend/public/study-group-hero.png` (the image supplied with the request).
- Displayed in the right column of the existing blue-gradient home hero.
- Cropped with `object-fit: cover`, rounded corners, and a subtle border/shadow;

the image keeps its source aspect ratio. On narrow screens the existing hero stacks into one column.

- Accessible image description: "Students studying together outdoors."
- The existing "Verified resources" badge overlays the photo. The existing

headline, text, navigation, and overall color system remain unchanged.

- "Explore documents" navigates to the dedicated `/library` page. The home page

does not render the Document Library section or send document-list requests.

- The public navigation still includes its separate “Document Library” link

to `/library`.

4. Student interfaces

Students use the signed-in workspace shell unless a row identifies a shared public interface.

Order	Route	Interface and exact contents
1	`/student-dashboard` and signed-in `/`	**Student learning overview.** Personalized welcome and Explore Documents action; recent submission, quiz-attempt, and account-role summaries; Quick Access cards for My Courses, Browse Documents, Practice Quizzes, and Upload Materials; recent submissions and recent quiz attempts with links to their full histories.
2	`/student/courses`	**My courses.** Course catalogue with course, year, semester, lecturer, enrollment/progress information, and Enroll/Leave course action; enrolled courses also link to their library and quizzes.
3	`/library`, `/courses/...`	**Document Library.** Search, folder navigation, filters, verified document cards, view/download actions, verification link when available, and signed-in report-document action.
4	`/quizzes`	**Published quiz catalogue.** Shared catalogue described above.
5	`/quizzes/:quizId`	**Quiz attempt.** Course/title header, question position and progress bar, one question with selectable answer options, answered count, Previous/Next, and Submit Quiz when every question is answered. Submission requires confirmation.
6	`/quizzes/:quizId` after submission	**Quiz result.** “Your Results,” quiz and course, server-calculated score, Try Again, and Back to Dashboard.
7	`/quiz-history`	**Quiz history.** Paginated attempts with quiz title, date, question count, and score; empty/loading/error states and a link back to quizzes.
8	`/document-history`	**Document submissions.** Upload action; submissions list with status filter, file/course/type/date, review note, verified-document and verification links where applicable, pagination, and empty/loading/error states.
9	`/upload`	**My uploads.** Course and document-type selectors, accepted file selection (PDF/JPEG/PNG/WebP; 10 MiB limit), file summary/removal, upload action, validation/submission feedback, and embedded upload-history panel. If student uploads are disabled, the upload form is replaced with the paused notice while history remains available.
10	`/verify` and `/verify/:token`	**Document verification.** Shared verification form and result states described above.

5. Lecturer interfaces

Lecturers use the signed-in workspace shell. The teaching overview scopes its course and performance information to courses assigned to the lecturer.

Order	Route	Interface and exact contents
1	`/lecturer` and signed-in `/`	**Teaching overview.** Review, quiz, assigned-course, student-attempt, and average-score summaries; assigned-course cards and actions for roster/progress, quiz reports, course-material upload, quiz management, and course announcements; learning-insights list for challenging questions.
2	`/staff`	**Review workspace.** Pending document cards with file preview, optional student-visible review note, Approve and Reject actions, and review-queue pagination.
3	`/quiz-management` and `/quiz-management?course_id=:id`	**Quiz management.** Paginated quiz list and create/edit form states; course/title/type/publication controls, question and answer-option editor, correct-answer selection, and save/delete actions. Lecturer course choices are limited to assigned courses.
4	`/quiz-results/:quizId`	**Quiz performance report.** Attempt count, average score, question-level correct-response/answer-choice breakdown, student attempts, and pagination.

Order	Route	Interface and exact contents
5	`/lecturer/courses/:courseId/students`	**Course student progress.** Enrollment, published-quiz, and page summaries; expandable student roster with enrollment date, quiz attempts, scores, and last activity; pagination.
6	`/lecturer/announcements?course_id=:id`	**Course announcements.** Course selector, publish/edit title and message form, announcements list with author/time, and edit/delete actions.
7	`/upload?course_id=:id`	**Course materials.** Shared upload interface initialized for the selected course; lecturers can publish to assigned courses and remove their verified materials from the library through submission history.
8	`/library`, `/courses/...`, `/quizzes`, `/verify`	**Shared library, quiz catalogue, and verification interfaces.** These use the same content and controls described in the public/student sections, inside the signed-in shell.

6. Moderator and review interfaces

Order	Route	Interface and exact contents
1	`/staff` and signed-in `/` for Moderator	**Moderation center.** Review queue plus moderation overview counts and report status shortcuts; report-management section with status filter, document/report details, moderator note, status actions, verified-file access, and pagination.
2	`/library`, `/courses/...`	**Document Library.** Shared searchable/filterable library inside the workspace shell.
3	`/verify` and `/verify/:token`	**Document verification.** Shared public verification interface inside the workspace shell.

Lecturers also use `/staff`, but see the review queue without moderator-only report-management controls. Administrators can use the same review/report interface.

7. Administrator interfaces

Order	Route	Interface and exact contents
1	`/admin` and signed-in `/`	**Platform overview.** Account, course, awaiting-review, and open-report totals; administrator responsibility cards; platform management links; role-scope cards.
2	`/admin/accounts`	**Account management.** Create account tab with name, email, initial password, and role; View accounts tab with role editing, save, delete (except current account), and refresh.
3	`/admin/courses`	**Course management.** Create course tab with code/title/year/semester/lecturer; View courses tab with lecturer assignment, delete, and refresh.
4	`/admin/settings`	**System settings.** Announcement enable/message, support contact, public registration toggle, student uploads toggle, and Save settings with success/error feedback.
5	`/admin/activity`	**Activity log.** Paginated account, course, document-review, and settings activity with actor, role, time, and load/error/empty states.
6	`/staff`	**Review and reports.** Review queue, and the report-management interface available to administrators for report filtering, notes, status updates, file access, and moderation actions.
7	`/quiz-management` and `/quiz-results/:quizId`	**Quiz management and performance report.** Shared administrator quiz creation/editing and performance-report interfaces.
8	`/library`, `/courses/...`, `/upload`, `/lecturer/announcements`	**Shared library, upload, and course-announcement interfaces.** Administrators can use the shared upload interface and manage announcements for available courses.

8. Route behavior and access

- When signed out, `/` displays the public home. When signed in, `/` redirects

to the user's role landing page: Student `/student-dashboard`, Lecturer `/lecturer`, Moderator `/staff`, Admin `/admin`.

- Signed-in visitors who open `/login` or `/register` are redirected to the

role landing page.

- `RequireAuth` protects student, lecturer, moderator, and administrator

destinations. A user without the required role is redirected according to the existing route guard.

- `/quizzes` is the published catalogue; `/quizzes/:quizId` is the protected

student attempt flow.

- `/courses/...` folder depth changes the selected library folder; it does not

introduce a separate visual page.

- Public and signed-in views of `/about`, the library, and verification share

the same page content but retain their respective shells.

- The frontend does not define separate password-reset, user-profile, or

standalone quiz-completion routes.

9. Shared interface states and behaviors

Data-backed pages expose their implemented loading, empty, and error states. Where the page provides retry, the retry action remains visible with the error. Mutations provide inline success/error feedback. Lists use their existing pagination controls. The following overlays/dialogs are part of the interface:

- Document report dialog from a library document card.
- Quiz submission confirmation before final submission.
- Native confirmation dialogs for destructive enrollment, document, course,

account, quiz, and announcement actions.

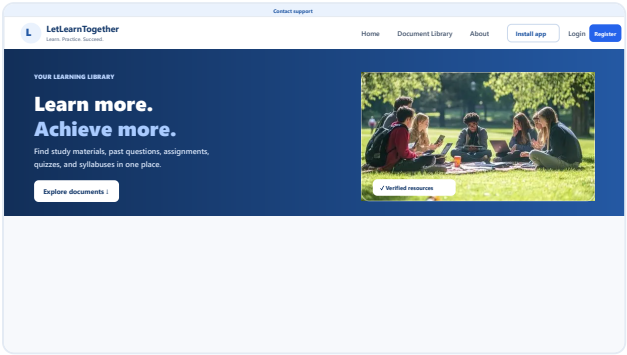
Forms retain their existing field labels, validation, disabled/loading button labels, and settings-dependent states. These states should be included when producing screenshots or presenting the interfaces; they are part of the implemented frontend, not additional pages.

10. Change-control checklist

Before changing a screen, check its route in `frontend/src/App.jsx`, its navigation entry, the shared shell, and the page component/CSS. After a visual change, update the matching route and visual details here in the same change. Do not add interface elements to this document unless they are present in the running application.

Public interface designs

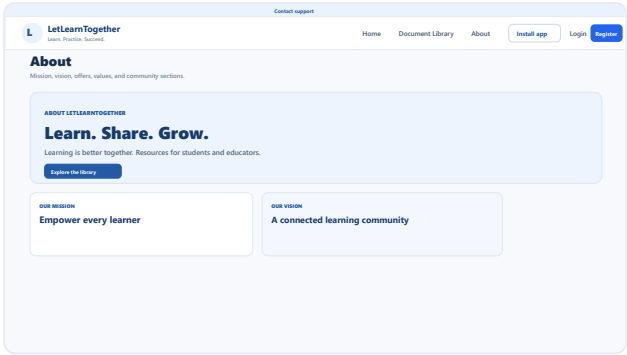
Illustrations use the implemented routes, controls, brand colors, and page structure. Repeated routes share the interface shown.



Home hero

`/`

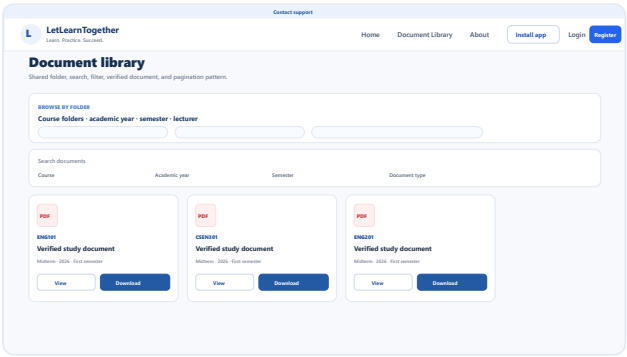
Hero-only home layout matching the current frontend.



About

`/about`

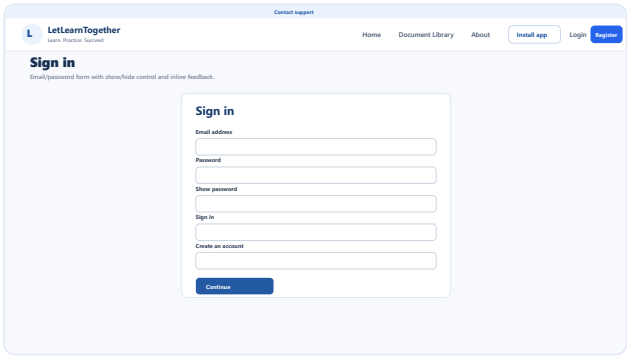
Mission, vision, offers, values, and community sections.



Document library

`/library and /courses/…`

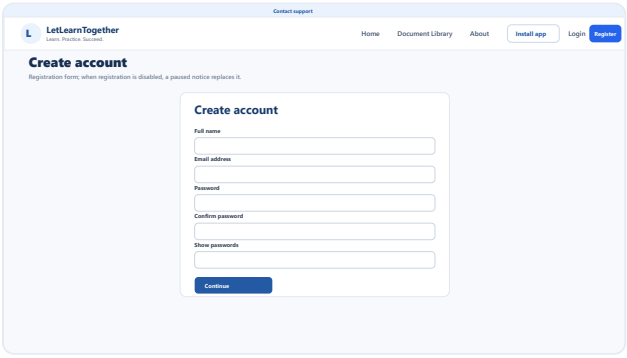
Shared folder, search, filter, verified document, and pagination pattern.



Sign in

`/login`

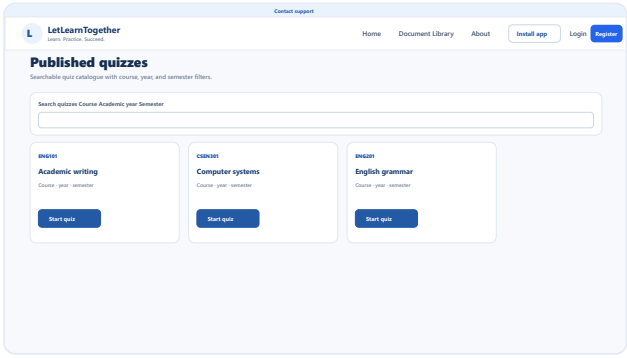
Email/password form with show/hide control and inline feedback.



Create account

`/register`

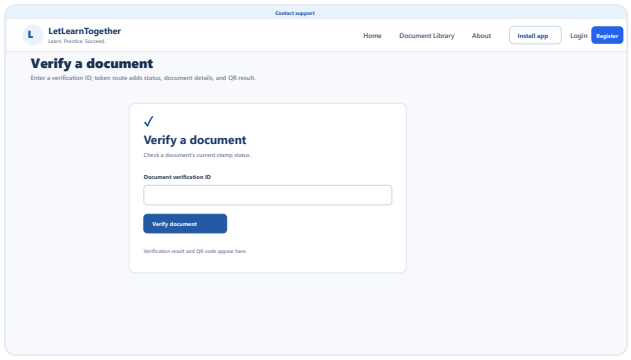
Registration form; when registration is disabled, a paused notice replaces it.



Published quizzes

`/quizzes`

Searchable quiz catalogue with course, year, and semester filters.



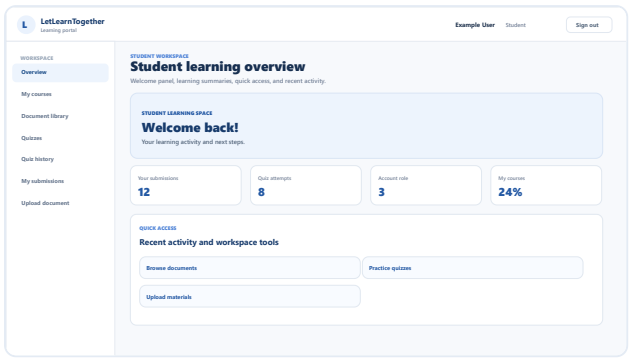
Verify a document

`/verify and /verify/:token`

Enter a verification ID; token route adds status, document details, and QR result.

Student interface designs

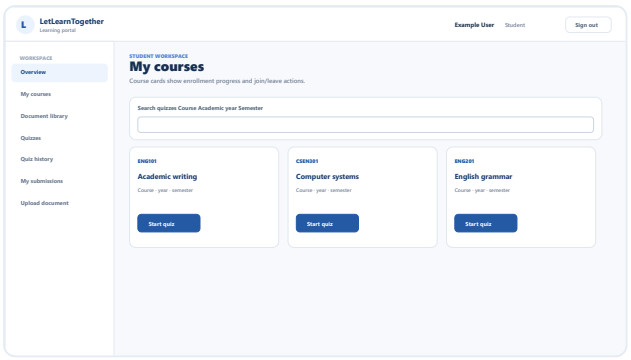
Illustrations use the implemented routes, controls, brand colors, and page structure. Repeated routes share the interface shown.



Student learning overview

`/student-dashboard (also signed-in /)`

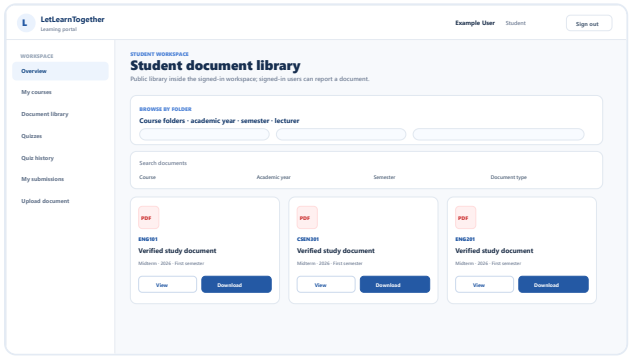
Welcome panel, learning summaries, quick access, and recent activity.



My courses

`/student/courses`

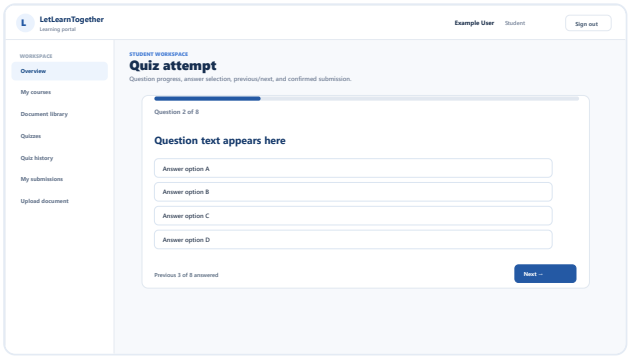
Course cards show enrollment progress and join/leave actions.



Student document library

`/library and /courses/…`

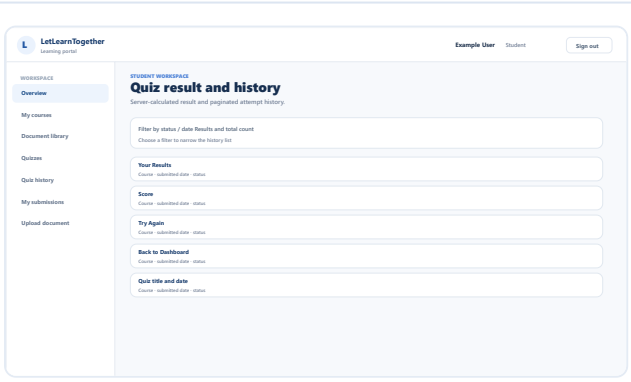
Public library inside the signed-in workspace; signed-in users can report a document.



Quiz attempt

`/quizzes/:quizId`

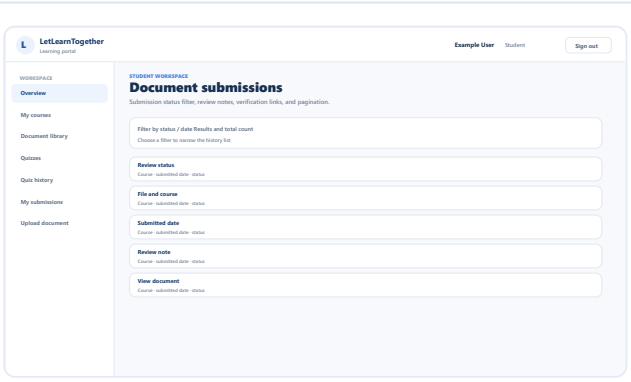
Question progress, answer selection, previous/next, and confirmed submission.



Quiz result and history

`/quizzes/:quizId after submit; /quiz-history`

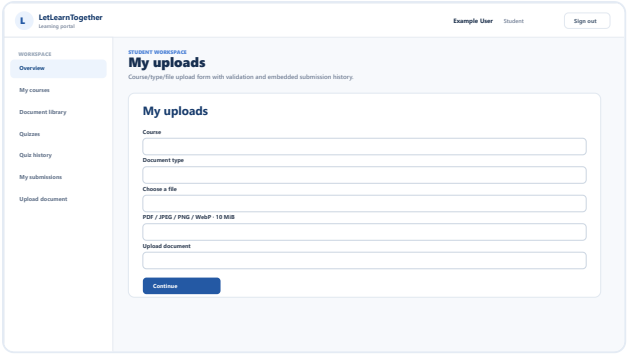
Server-calculated result and paginated attempt history.



Document submissions

`/document-history`

Submission status filter, review notes, verification links, and pagination.



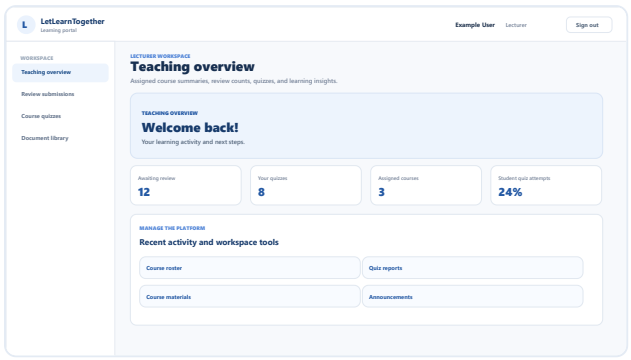
My uploads

`/upload`

Course/type/file upload form with validation and embedded submission history.

Lecturer interface designs

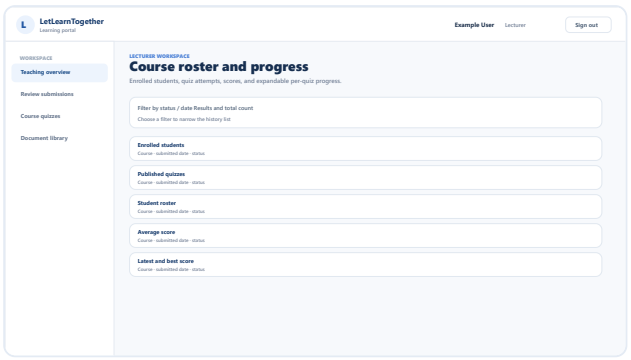
Illustrations use the implemented routes, controls, brand colors, and page structure. Repeated routes share the interface shown.



Teaching overview

`/lecturer (also signed-in /)`

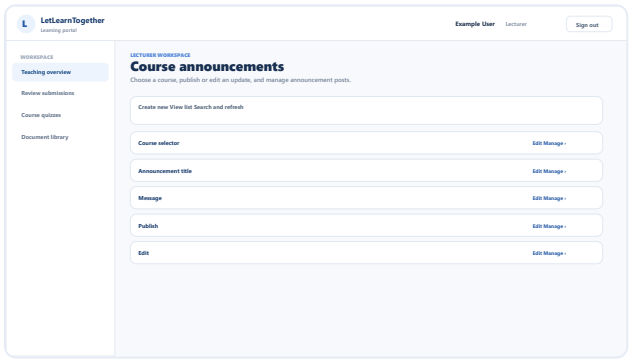
Assigned course summaries, review counts, quizzes, and learning insights.



Course roster and progress

`/lecturer/courses/:courseId/students`

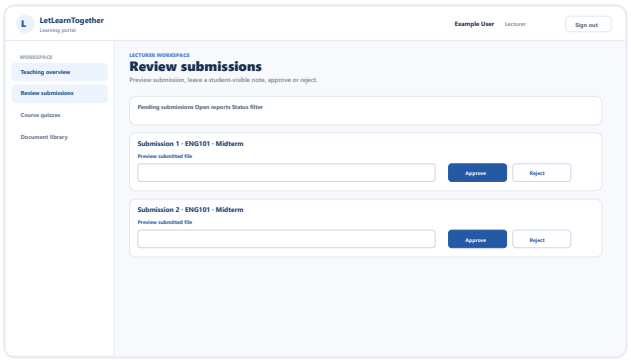
Enrolled students, quiz attempts, scores, and expandable per-quiz progress.



Course announcements

`/lecturer/announcements?course_id=:id`

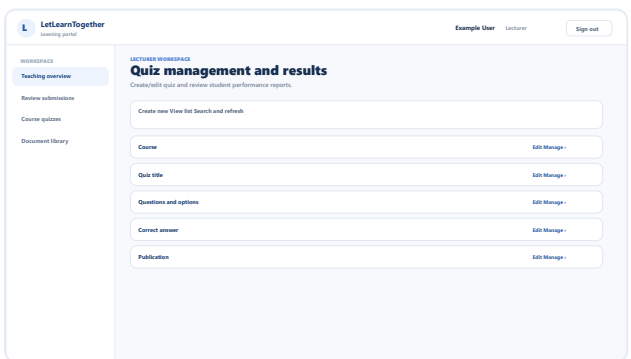
Choose a course, publish or edit an update, and manage announcement posts.



Review submissions

`/staff`

Preview submission, leave a student-visible note, approve or reject.



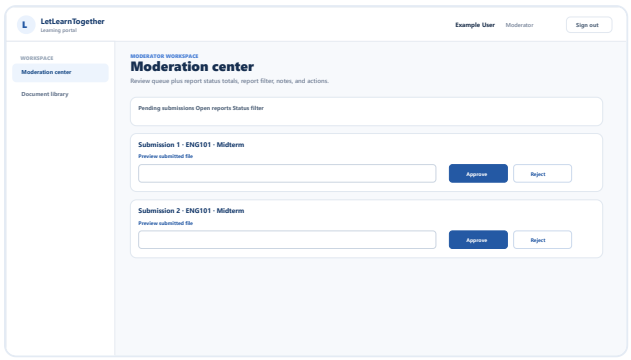
Quiz management and results

`/quiz-management and /quiz-results/:quizId`

Create/edit quiz and review student performance reports.

Moderator interface designs

Illustrations use the implemented routes, controls, brand colors, and page structure. Repeated routes share the interface shown.



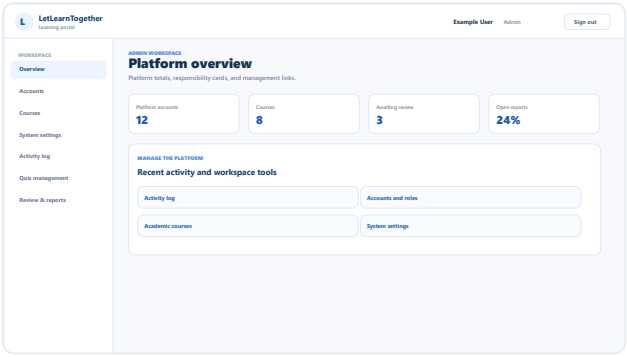
Moderation center

`/staff and signed-in /`

Review queue plus report status totals, report filter, notes, and actions.

Admin interface designs

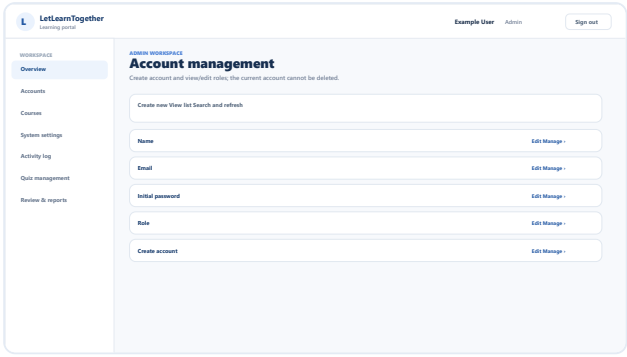
Illustrations use the implemented routes, controls, brand colors, and page structure. Repeated routes share the interface shown.



Platform overview

`/admin (also signed-in /)`

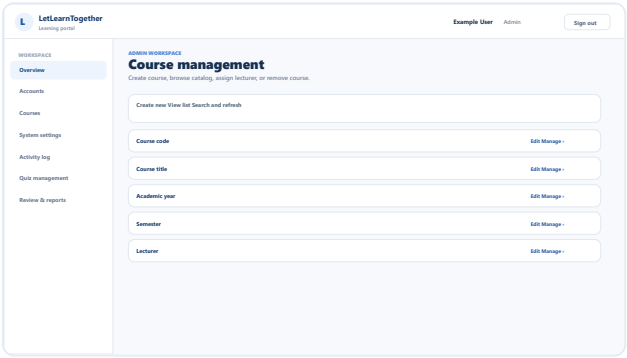
Platform totals, responsibility cards, and management links.



Account management

`/admin/accounts`

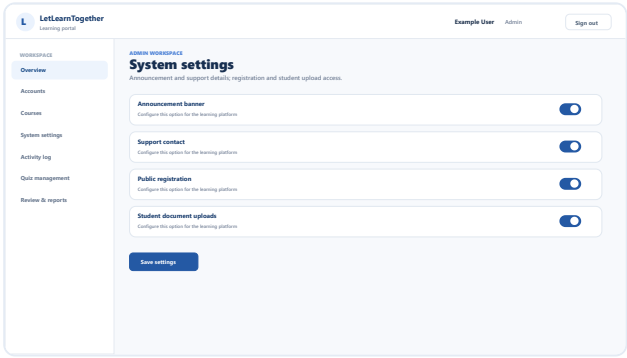
Create account and view/edit roles; the current account cannot be deleted.



Course management

`/admin/courses`

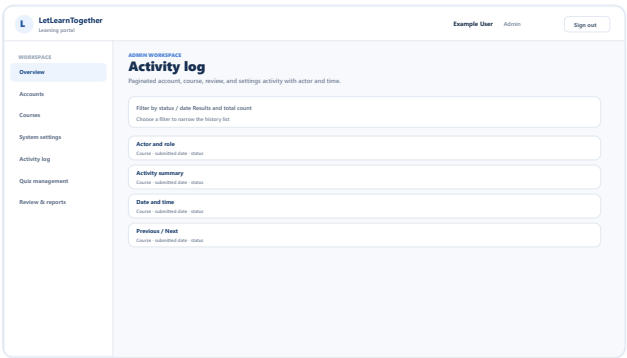
Create course, browse catalog, assign lecturer, or remove course.



System settings

`/admin/settings`

Announcement and support details; registration and student upload access.



Activity log

`/admin/activity`

Paginated account, course, review, and settings activity with actor and time.